CAN THEY BE USED INTERCHANGEABLY?

knowledgeiswatt.substack.com/p/62-flat-vs-uphill-power-output

Knowledgeiswatt February 27, 2025

Photo by Diana Rafira on Unsplash

It is common feeling easier to produce watts when riding uphill compared to flat terrain.

Some coaches even recommend slightly higher power values for a given training zone when riding uphill compared to flat. Is it right? Does this adjustment depend on the riders being a climber or a flat specialists?

A study published by Valenzuela et al. in 2022 on *Journal of Science and Medicine in Sport* tried to answer this question. (1)

WHAT DID THEY DO?

- They retrospectively analysed the maximal mean power outputs (MMPO) in training and races over fixed duration (1min, 5min, 10min, 20min) of 98 male road pro cyclists from 2 different teams (1 World Tour and 1 Pro Tour) between 2013 and 2022.
- The MMPOs were divided into:
  - achieved on flat terrain: if average road slope was < 5%
  - achieved uphill: if average road slope was > 5%

WHAT DID THEY FIND?

1. **Higher MMPOs (from ~+0.5 to +4% for all the efforts' duration) were found uphill compared to flat terrain.**
2. **A tendency for a larger improvements when passing from flat to uphill terrain for 10min and 20min MMPOs was observed for climbers vs flat specialists:**
   - climbers: +3.4 ± 3.2 % for 10min MMPO and +3.9±3.2 % for 20min MMPO;
   - flat specialists: +1.9 ± 3.6 % for 10min MMPO and +1.6±5.1 % for 20min MMPO;
3. **Almost all the best RPOs were observed at 5-10% slope, with an average slope of 6-7%.**

PRACTICAL TAKEAWAYS

- **6-7% slope is the optimal road gradient to obtain maximal power output over 1-20 min for both climbers and flat specialists.**
- **Climbers tends to increase more their MMPOs when passing from flat to uphill terrain.**

GABRIELE'S COMMENT

How can we use these data?

- When doing tests and comparing performance, keep in mind that mean maximal power outputs for efforts of different duration are obtained at 6-7% of road gradient. So, if performing a 20min test or race efforts on flat expect the power values to be lower compared to uphill.

- **Training: if training zones were calculated from a test on a climb, consider reducing by ~3% power output on flat terrain. Conversely, if you performed a test on a flat terrain, increase by ~3% power output when doing uphill efforts.**

  This holds true particularly for climbers.

Why these flat vs uphill power output differences were observed?

We don’t have a definitive answer, but it could be due the position: standing on the pedals on the climbs makes possible to push more watts, while this position on the flat is avoided because of the aerodynamics disadvantages outperform the possible power output advantages.

Thanks for reading! Spread with your team!

Gabriele Gallo, PhD in Exercise and Sport Science

REFERENCES:

---

1. Valenzuela PL, Mateo-March M, Muriel X, Zabala M, Lucia A, Pallares JG, Barranco-Gil D. Road gradient and cycling power: An observational study in male professional cyclists. J Sci Med Sport. 2022 Dec;25(12):1017-1022.